

ESP32-IDF Configuration Tools

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Note: The out-of-the-box virtual machine already includes the environment, so there is no need to download it again.

1. ESP32-IDF Configuration Tools Introduction

In the root directory of an ESP32-IDF project, there is an sdkconfig file that is used to store the ESP32-IDF configuration content.

2. Activating the ESP-IDF Development Environment

Run the following command in the esp-idf tools directory.

```
source ~/esp/esp-idf/export.sh
```

Note: Every time you open a new terminal, you must first activate the ESP-IDF development environment before you can compile ESP-IDF projects.

3. Creating a New Project

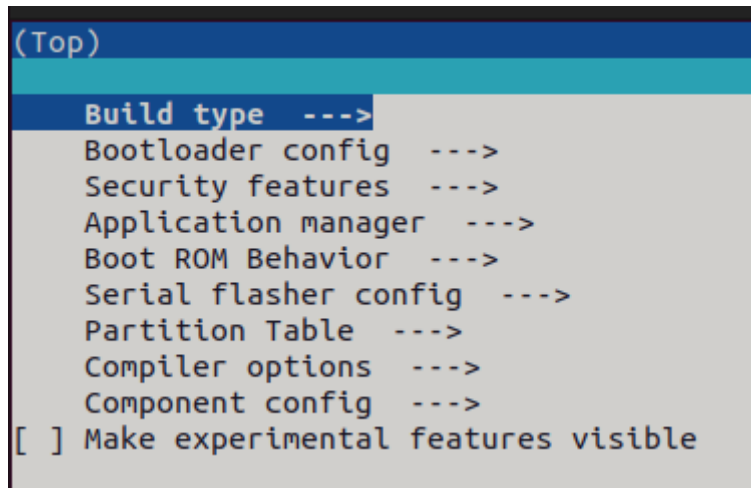
Create a new blank project to use as a base project.

```
mkdir ~/esp/samples/esp32_samples  
cd ~/esp/samples/esp32_samples  
idf.py create-project main  
cd main
```

4. Configuring the Project

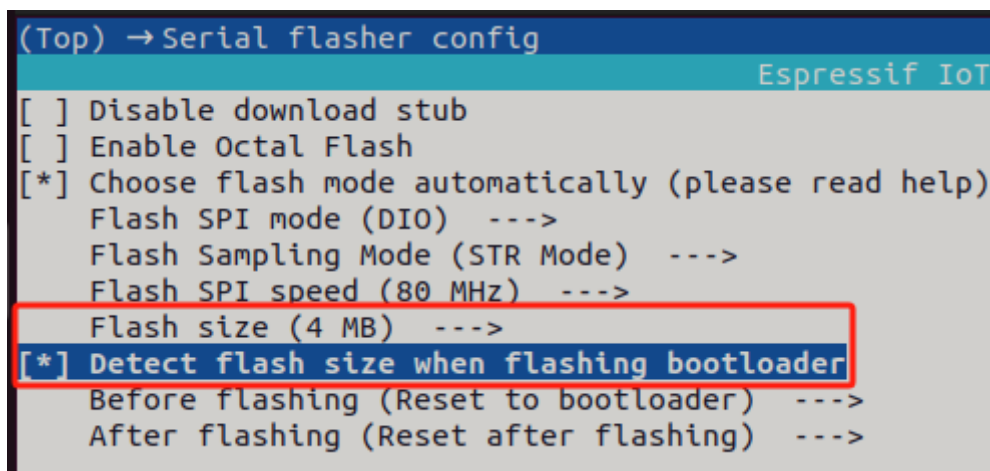
Set the target chip to the esp32s3 chip and open the configuration interface.

```
idf.py set-target esp32s3  
idf.py menuconfig
```



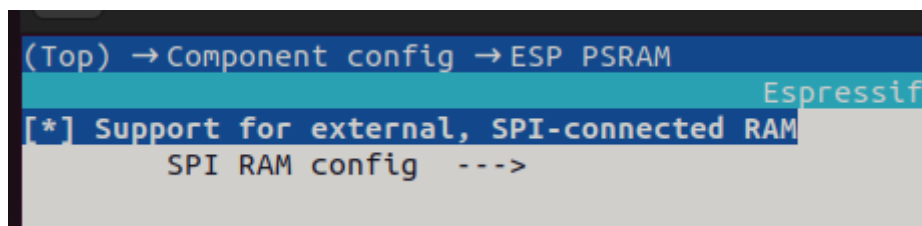
1. Configure the automatic detection of the flash size during flashing.

In Serial flasher config, change Flash size to 4MB, and check Detect flash size when flashing bootloader.

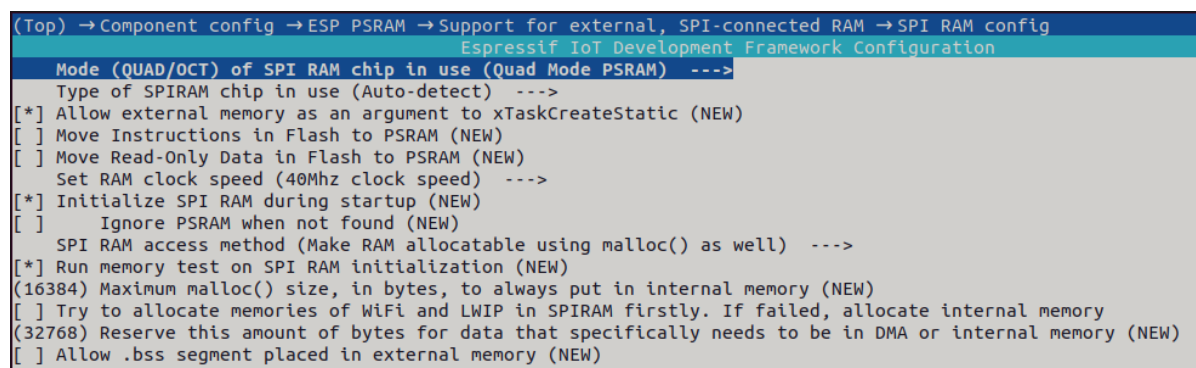


2. Configure the external PSRAM

Open Component config->ESP PSRAM, and enable Support for external, SPI-connected RAM.

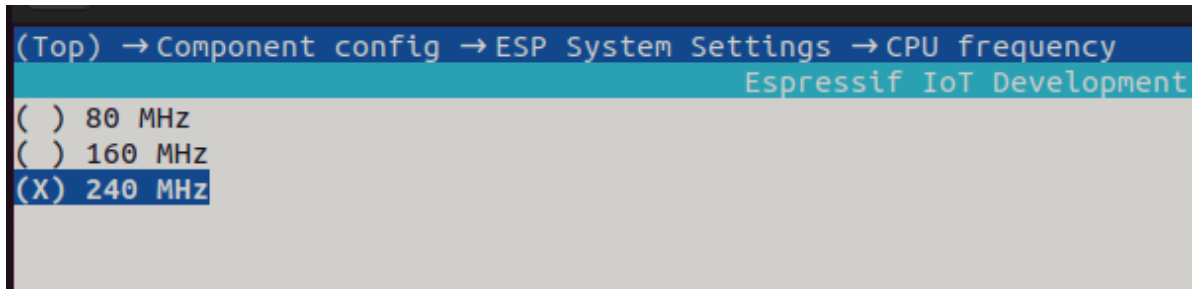


In the SPI RAM config configuration, keep the defaults as shown in the figure below.



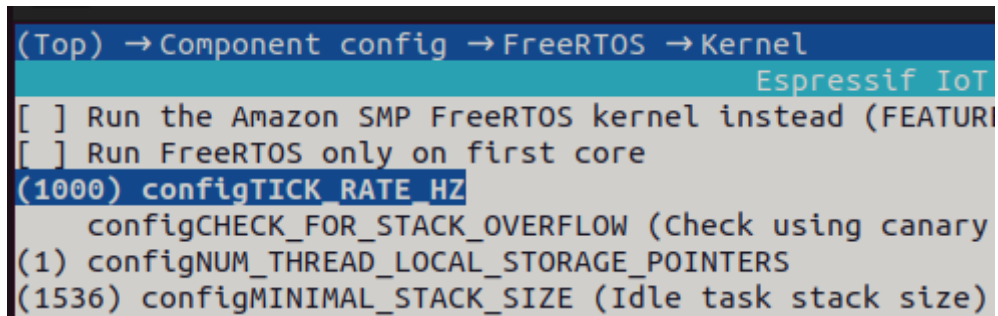
3. Configure the CPU frequency to 240MHz

Open Component config->ESP System Settings->CPU frequency, and change the CPU frequency to 240MHz.



4. Configure the FreeRTOS tick frequency to 1000Hz

Open Component config->FreeRTOS->Kernel, find configTICK_RATE_HZ, and change the value to 1000.



5. Configure the partition table to increase the program space

Open Parttition Table->Partition Table, and select Single factory app (large), no OTA.

